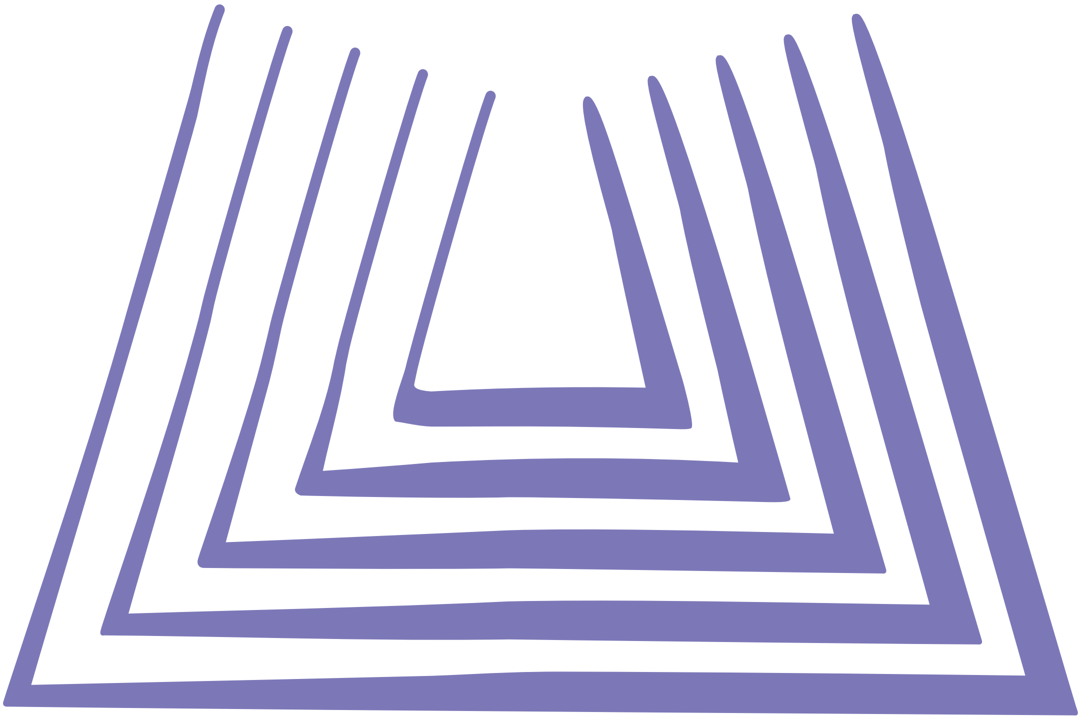
Compendium of WHO and other UN guidance on health and environment

World Health Organization

Chapter 5. Chemicals



UNDP

UN environment

unicef for every child

WHO/HEP/ECH/EHD/21.02

Suggested citation. Chemicals. In: Compendium of WHO and other UN guidance on health and environment. Geneva: World Health Organization; 2021 (WHO/HEP/ECH/EHD/21.02)

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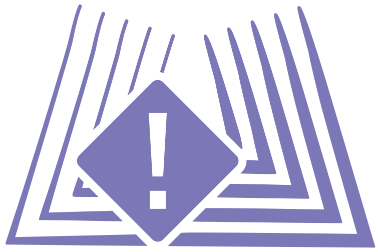
5.1 Introduction

More than 160 million chemicals are known to humans *(1)*. About 40 000 to 60 000 of them can be found in commerce; 6000 of these account for more than 99% of the total volume of chemicals in commerce globally *(2)*. In 2017, the chemical industry was the second largest manufacturing industry in the world and the trend is going upwards – chemicals sales are projected to almost double from 2017 to 2030 *(3)*.

Chemicals, whether of natural origin or produced by human activities, are part of our environment. Manufactured chemicals include industrial and agricultural products such as pesticides, petroleum products and processed metals. Some chemicals are manufactured for specific uses, while others are unwanted by-products, including wastes, or products of combustion such as toxic gases and particles from industrial emissions and burning of fuel.

All people come in contact with chemicals as part of normal life – through the food and drinking-water they consume, the products they use or are surrounded by at home or the workplace, through the contact with the environment (e.g. through breathing air, touching the soil, and swimming in recreational waters) or as a result of a chemical incident. Many of the chemicals people use and are exposed to are harmless or even beneficial; others pose a threat and are hazardous to people's health and the environment. Levels of exposure and resulting health impacts are determined by social as well as biological factors. Men, women and children are exposed to different kinds and levels of chemicals and are exposed with different frequency. In addition, men, women and children vary in their physiological susceptibility to health effects from exposure to hazardous chemicals *(4)*.

5.2 Chemical safety



In May 2017, the Seventieth World Health Assembly approved the *Chemicals road map* to enhance health sector engagement in the Strategic Approach to International Chemicals Management (SAICM) *(5)*. The road map identifies concrete actions where the health sector in countries has a lead or important supporting role to play in the sound management of chemicals, recognizing the need for multisectoral cooperation. An important role of the health sector is to make a contribution to reducing health risks from exposures to chemicals by promoting health protection strategies, regulating chemicals, increasing public education, and sharing information and best practices. In doing so, the role of the health sector is to increase knowledge and evidence about the toxicological properties of chemicals and related human health risks and impacts. Another role of the health sector is to promote the inclusion of health considerations in all chemicals policies including those developed by other sectors.

At the national level, countries usually have in place laws to ensure the safe handling of chemicals and to protect the environment, consumers and workers from contamination by and exposure to hazardous chemicals. In addition, laws are in place to prevent, prepare and respond to chemical incidents, including accidents at hazardous installations (e.g. chemical plants) and during transport. Regulations specify classification, labelling, packaging and transport of hazardous materials, including hazardous chemicals *(6)*. National laws also define chemical emission and quality standards, for example permitted concentration of chemicals in air, water, food and consumer products. Specific laws regulate the management of groups of chemicals, for example the management of pesticides.

Overview

In 2016, a small number of chemicals for which data are available were estimated to cause 1.6 million deaths from a variety of health outcomes including poisonings, heart diseases, chronic respiratory diseases and cancers *(7)*. Chemical pollution also negatively impacts a range of facets of the ecosystem, which can harm human health.

Some hazardous chemicals are of particular health concern because of their widespread presence in the environment, their toxicity and capacity to magnify and accumulate in environmental and human media, and the fact that many humans easily come in contact with them thereby harming the health of large populations. Chemicals or groups of chemicals of major public health concern include air pollution, arsenic, asbestos, benzene, cadmium, dioxin and dioxin-like substances, inadequate or excess fluoride, lead, mercury and highly hazardous pesticides (HHPs) *(8)*.

Who is impacted by unsafe levels of chemicals in my country?	**Exposure of the general population** Some countries routinely conduct population surveys to study the status and trend of chemical exposures (e.g. through the environment, food, and consumer products) and the related health risks of (sub-) populations. Often chemical exposure is determined by human bio-monitoring, that is, concentrations of chemicals are measured in human fluids (e.g. blood, urine) or tissues (e.g. hair, fingernail) *(9, 10)*. **Exposure through air, water and food** Mandatory environmental and food monitoring programmes routinely measure chemicals in certain contexts, for example ambient and indoor air, surface and groundwater, and various food items, as well as in occupational environments. Often these programmes focus on monitoring specific chemicals indicating a broader exposure pattern and, therefore, the range of substances being monitored may be limited (see Chapter 2 Air pollution and sections 3.2.1 Drinking-water, 10.1 Food safety and the environment, and 11.3 Workplaces). Environmental monitoring data provide an estimate of the health risks when compared to the WHO air quality guidelines 2005 update and WHO GDWQ *(11, 12)*. Food monitoring data of pesticide residues, food additives and contaminates can be compared with the guidance values (e.g. Acceptable Daily Intakes prepared by the Joint FAO/WHO Expert Committee on Food Additives and the Joint FAO/WHO Meeting on Pesticide Residues *(13-15)*. **Exposure through soil** In soils, chemicals are generally only assessed when contamination is suspected, for example in the case of an abandoned waste site where there is a pollution risk to groundwater. Activities and industries that have shown to pollute soil include used lead acid battery recycling, mining and ore processing, tanneries, dumpsites, industrial estates, smelting, artisanal small-scale gold mining, product manufacturing, chemical manufacturing and the dye industry.
What levels of chemicals in the air, water, products, etc. do we want to achieve?	*Chemicals in air:* WHO air quality guidelines are available for a number of pollutants and are presented in concentrations of pollutants in the air: particulate matter, ozone, nitrogen dioxide and sulfur dioxide in ambient air, and additional chemicals in indoor air *(12, 16)* (see Chapter 2 Air pollution). *Chemicals in drinking-water:* WHO drinking-water quality guidelines propose guideline values for a wide variety of chemicals *(11)*. *Chemicals in soil:* Some countries have set standards for contaminants in soils in residential areas and for farming and crop production. *Chemicals in food:* WHO and the Food and Agriculture Organization of the United Nations (FAO) implement the Codex Alimentarius, which is a collection of standards, guidelines and codes of practice on foods and their contents, including chemical contaminants and food additives *(15, 17, 18)*. Additional guidance and guideline values (e.g. occupational exposure limits) can be found in Environmental Health Criteria Documents, Concise International Chemical Assessment or International Chemical Safety Cards (ICSCs) – all are available in the International Programme on Chemical Safety (INCHEM) database *(9)* and in the Inter-Organization Programme for the Sound Management of Chemicals (IOMC) Internet-based Toolbox for Decision Making in Chemicals Management (IOMC Toolbox) *(19)*.

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
Policies and actions			
1. Implement the WHO *Chemicals road map* to enhance health sector engagement in the SAICM towards the 2020 goal and beyond, approved by the World Health Assembly in 2017 *(5)*. • Manage health risks from exposures to chemicals: Develop health protection strategies, regulate chemicals, educate the public about health risks, share information and best practices, develop high-quality health care settings. • Improve knowledge and evidence about the health effects and impacts of chemicals: improve risk assessment methodologies, increase bio-monitoring and surveillance, estimate disease burden from chemicals, share information and collaborate with partners. • Strengthen national capacities to address health threats from chemicals, including in response to chemical incidents and emergencies: Strengthen national policies and regulatory frameworks, implement the International Health Regulations (2005) and provide training and education. • Strengthen leadership and coordination: Promote the inclusion of health considerations in all policies related to chemicals, engage the health sector in chemicals management activities at the national, regional and international levels, and engagement of the health sector with other sectors.	Health	National; healthcare; workplace	Governance; regulation; assessment and surveillance; information, education and communication
Note: Some of the below recommendations (2–9, 11 and 12) are actions included in the WHO *Chemicals road map*.			
2. Implement the International Health Regulations (2005) to establish/strengthen core capacities for chemical incident and emergency preparedness, detection and response chemical events, including poison centre and laboratory capacities *(20)*.	Health Environment Labour Other sectors	National	Regulation
3. Implement the chemicals and waste-related multilateral environmental agreements, particularly health protective aspects, e.g.: • Minamata Convention on Mercury *(21)* • Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal *(22)* • Rotterdam Convention on Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade *(23)* • Stockholm Convention on Persistent Organic Pollutants (POPs) *(24)* • Montreal Protocol on Substances that Deplete the Ozone Layer *(25)*.	Health, Environment Agriculture Other sectors	National	Regulation
4. Nominate a health ministry contact point for the WHO Global Chemicals and Health Network *(5)*.	Health	National	Governance
5. Support the inclusion of health priorities in all policies relevant to chemicals *(5)*.	Health Other sectors	National	Governance

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
6. Facilitate participation of all relevant sectors and stakeholders in chemicals management and strengthen the engagement of the health sector with other sectors, recognizing the shared leadership of the health and environmental sectors *(5)*.	Health Environment Agriculture Labour Other sectors	National	Governance
7. Establish health-based guidelines for chemicals in water, air, soil, food, products, and occupational exposure, drawing on WHO norms, standards and guidelines, as appropriate, and participating in their development *(5)*.	Health Other sectors	National	Regulation
8. Support regulations to prevent discharge of toxic chemicals and advocate appropriate recovery and recycling technology, as well as safe storage and disposal *(5)*.	Health Other sectors	National	Regulation
9. Support implementation of the Globally Harmonized System of Classification and Labelling of Chemicals, coordinating internationally, where appropriate *(5, 6)*.	Health Other sectors	National	Regulation
10. Prevent the construction of homes, schools and playgrounds near polluted areas and hazardous installations *(26, 27)*.	Health Land use planning Construction Housing	National; community	Regulation
Awareness raising and capacity building			
11. Conduct public awareness campaigns for priority health concerns related to chemicals throughout their life cycle (e.g. HHPs, lead, mercury, etc.) *(5)*.	Health	National; community Universal health coverage	Information, education and communication
12. Promote communication of relevant information, including training, on chemicals used in products and processes, to enable informed decision-making by all actors throughout a product's life cycle, and to promote safer alternatives *(5)*.	Health Labour Other sectors	National; community; workplace	Information, education and communication
13. Educate and raise awareness about the health effects of chemicals and about actions to prevent exposure to toxic chemicals *(28, 29)*.	Health	National; community; workplace Universal health coverage	Information, education and communication

Guidance	Sector principally involved in planning/implementation	Level of implementation	Instruments
14. Promote safe storage of chemicals at home *(26, 27)*: • Keep all chemicals and medicines out of the reach of children, either locked or stored in places they cannot access. This applies to cleaning products in kitchen and bathroom, paraffin or kerosene, medicines, fuels and caustic products in the garage or pesticides in the shed. • Chemicals should never be stored in drinking bottles.	Health	National; community Universal health coverage	Information, education and communication
15. Promote the use of child-resistant packages for pharmaceuticals and for chemical products *(26)*.	Health	National; community Universal health coverage	Information, education and communication
16. Ensure clear labelling for cleaners, fuels, solvents, pesticides and other chemicals used at home and in schools *(27)*.	Health	National; community Universal health coverage	Information, education and communication
17. Inform parents, teachers and child-minders about the potential chemical hazards in the places where children spend their time *(26)*.	Health	National; community Universal health coverage	Information, education and communication
18. Raise awareness among families and communities about poison control centres *(26)*. (See also the *World directory of poison centres (30)*).	Health	National; community Universal health coverage	Information, education and communication

Specific actions on priority chemicals of concern – These include the ten chemicals of major public health concern as listed by INCHEM *(8)*. There are various other chemicals, including persistent organic pollutants, which are not included here but are highly hazardous and negatively affect health and the environment, especially when improperly managed.

Arsenic – Reduction of arsenic exposure through drinking-water *(31)*			
19. Screen drinking-water and identify where delivered water is above the WHO provisional guideline value of 10 µg-arsenic per litre or national permissible limit. Combine screening activities with awareness-raising campaigns.	Health Environment Water	National; community	Assessment and surveillance
20. Use alternative groundwater sources, microbiologically safe surface water (e.g. rainwater harvesting) or arsenic removal technologies.	Health Environment Water	National; community	Infrastructure, technology and built environment

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
Asbestos – Elimination of asbestos-related diseases *(32)*			
21. Stop the use of all types of asbestos as the most efficient way to eliminate asbestos-related disease.	Health Construction Housing	National; community	Regulation
22. Replace asbestos with safer substitutes and develop economic and technological mechanisms to stimulate its replacement.	Construction Housing	National; community	Infrastructure, technology and built environment; regulation
23. Take measures to prevent exposure to asbestos in place and during asbestos removal (abatement).	Construction Housing	National; community	Infrastructure, technology and built environment; regulation
24. Improve early diagnosis, treatment, social and medical rehabilitation of asbestos-related diseases and establish registries of people with past and/or current exposures to asbestos.	Health	Health care; community Universal health coverage	Assessment and surveillance; other management and control
Benzene – Interventions to reduce worker and population exposure *(33)*			
25. Support the use of alternative solvents in industrial processes.	Industry	Workplace	Regulation
26. Develop or update policies and legislation to remove benzene from consumer products and to discourage domestic use of benzene-containing products.	Industry Health	National	Regulation
27. Promote building codes requiring detached garages.	Construction Housing	National	Regulation
Cadmium – Interventions to reduce work and population exposure *(34)*			
28. Reduce cadmium emissions from mining and smelting, waste incineration, application of sewage sludge to the land, and use of phosphate fertilizers and cadmium-containing manure, among others.	Industry Agriculture Other sectors	Workplace; national; community	Infrastructure, technology and built environment
29. Support safe and effective measures to increase recycling of cadmium.	Industry Other sectors	Workplace; national; community	Infrastructure, technology and built environment
30. Restrict non-recyclable uses of cadmium.	Industry Other sectors	National	Regulation

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
31. Support the elimination of use of cadmium in products such as toys, jewellery and plastics.	Industry Health Other sectors	National	Other management and control
Dioxins and dioxin-like substances – Actions to reduce emissions of these substances (required by the Stockholm Convention) *(35)*			
32. Identify and safely dispose of material containing or likely to generate dioxin and dioxin-like substances such as electrical equipment.	Environment Other sectors	Workplace; national; community	Regulation; infrastructure, technology and built environment
33. Ensure appropriate combustion practices to prevent emissions of dioxins and dioxin-like substances.	Environment Waste Industry Other sectors	Workplace; national; community	Regulation; infrastructure, technology and built environment
34. Implement FAO/WHO strategies to reduce contamination in food and feed, and monitoring of food items and human milk *(36)*.	Food Agriculture	National	Regulation; assessment and surveillance
Inadequate or excess fluoride *(37)*			
35. Ensure sufficient fluoride intake where this is lacking, so as to minimize tooth decay.	Health Water	National	Regulation
36. Provide drinking-water with a moderate (i.e. safe) fluoride level in areas where groundwater contains high fluoride levels. Guideline values for fluoride in drinking-water and air are available *(37)*.	Water Health	National; community	Regulation
37. Provide guidance on the need to control population exposures to fluoride and establish the important balance between caries prevention and protection against adverse effects.	Health	National	Regulation
Lead – Risk mitigation recommendations *(38, 39)*			
38. Develop and enforce health, environmental and safety standards for manufacturing and recycling of lead-acid batteries, e-waste and other substances that contain lead *(40)*.	Health Industry Labour Environment Other sectors	National; community	Regulation

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
39. Enforce environmental and air-quality regulations for smelting operations.	Health Industry Labour Environment Other sectors	National; community	Regulation
40. Manage drinking-water safety so that quality standards have strict parameters on lead *(11)*.	Health Industry Labour Environment Other sectors	National; community	Regulation
41. Ensure that health care practitioners have training on, and resources for, the diagnosis and management of lead poisoning.	Health	National	Information, education and communication
42. Ensure the availability of laboratory capacity for blood lead testing.	Health	National; community	Regulation
43. Phase out the use of lead additives in fuels and lead in paint where this has not yet been done; adopt legally binding limits on lead in paint.	Industry Transport Environment Health	National	Regulation
44. Eliminate the use of leaded solder in food and drink cans and water pipes; lead in homes, schools, school materials and children's toys; lead glazing for pottery intended for cooking, eating or drinking; spices; and lead in traditional medicine and cosmetics.	Industry Other sectors	National; community	Regulation
45. Identify contaminated sites and exposure routes and take necessary action to prevent human exposure to lead from these areas. Identify sources of lead exposure in children such as lead in contaminated soil, paint, toys, water distribution pipes, etc. *(27)*.	Environment Health Other sectors	National; community	Assessment and surveillance
46. Monitor blood lead concentrations in populations at risk by sensitive analytical methods *(41)*.	Health	National; community Universal health coverage	Assessment and surveillance

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
47. Enhance the collection of data on lead in foodstuffs and make this information publicly available so that appropriate action can be taken.	Food	National; community	Assessment and surveillance
48. Educate the public regarding the dangers of misusing lead-containing products, which covers risks from lead exposure and the ways to protect themselves, their families and their communities. This may include public education campaigns to parents and caregivers, schools including classroom teachers and students, youth associations, community leaders and health care workers, workers and owners of lead-related industries (e.g. lead-acid battery recyclers and smelters, ceramic potters, spice adulterators). Existing media and communication resources and mediums may be used to reach audiences that may not be aware of the risks of lead exposure to children and pregnant women.	Health Education	National; community Universal health coverage	Information, education and communication
49. Promote preventive and educational measures to protect young children from lead in their environment.	Health Education	National; community Universal health coverage	Information, education and communication
Mercury – Interventions to prevent health risks from mercury exposure *(42, 43)*			
50. Implement the Minamata Convention on Mercury.	Environment Health	National	Regulation
HHPs – Measures to reduce exposures to HHPs and their health impacts *(38)*			
51. Establish national regulation of the registration, labelling, marketing, purchase and use of pesticides, including HHPs *(44-46)*.	Agriculture	National	Regulation
52. Implement the FAO guidance on the appropriate handling and use of pesticides *(47)*.	Agriculture	National	Regulation
53. Eliminate the use of persistent HHPs and inappropriate wastes, especially HHPs subject to the Stockholm and Rotterdam Conventions *(23, 24)*.	Agriculture	National	Regulation
54. Raise awareness and understanding among pesticide users about the importance and ways of protecting health and the environment from the possible adverse effects of pesticides and the existence of less hazardous alternatives.	Health Agriculture Environment	National; community Universal health coverage	Information, education and communication
55. Educate and inform health professionals on recognition and treatment of pesticide related poisoning *(38)*.	Health	Health care	Information, education and communication

Note: Actions for risk reduction from the use of chemicals in health care settings are listed in section 11.4 Health care facilities. Additional information and more comprehensive guidance on health sector engagement is available in the *Chemicals road map (5)*.

Selected tools

WHO 2017: *Chemicals road map and workbook (5)*

The road map identifies concrete actions where the health sector has a lead or important role to play in the sound management of chemicals.

The associated workbook helps to prioritize and plan actions outlined in the chemicals road map.

WHO 2020: *Ten chemicals of major public health concern (8)* includes information for decision-makers, including tools for action, norms and guidelines, fact sheets, etc. Further resources on the ten chemicals are below.

Air pollution
See Chapter 2 Air pollution

Arsenic
WHO/UNICEF 2018: *Arsenic primer: guidance on the investigation & mitigation of arsenic contamination (48)*

Asbestos
WHO 2014: *Chrysotile asbestos (49)*

Benzene
WHO 2019: *Exposure to benzene: a major public health concern (33)*

Cadmium
WHO 2019: *Exposure to cadmium: a major public health concern (34)*

Dioxin and dioxin-like substances
UNEP/Stockholm Convention 2013: *Toolkit for identification and quantification of releases of dioxins, furans and other unintentional POPs (50)*

Fluoride
WHO 2013: *Oral health surveys: basic methods – 5th ed. (51)*

Lead
WHO 2020: Global elimination of lead in paint: why and how countries should take action. Policy brief *(52)* and technical brief *(53)*

WHO 2020: *Brief guide to analytical methods for measuring lead in paint, 2nd edition (54)*

WHO 2020: *Brief guide to analytical methods for measuring lead in blood, 2nd edition (55)*

WHO 2020: Guidance on organizing an advocacy or awareness-raising campaign on lead paint *(56)*

UNEP 2018: *Model law and guidance for regulating lead paint (57)*

Mercury
WHO 2021: Minamata Convention on Mercury: annotated bibliography of WHO information *(43)*

WHO 2021: Exposure to mercury: a major public health concern, second edition *(58)*

WHO 2019: Addressing health when developing national action plans on artisanal and small-scale gold mining under the Minamata Convention on Mercury *(59)*

WHO 2019: Strategic planning for implementation of the health-related articles of the Minamata Convention on Mercury *(60)*

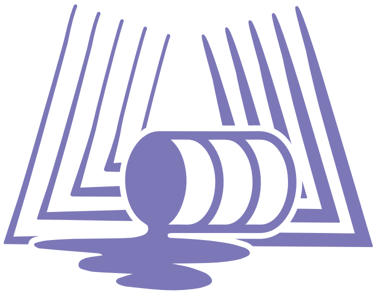
WHO 2018: Health sector involvement in the Minamata Convention on Mercury *(61)*

WHO 2015: Developing national strategies for phasing out mercury-containing thermometers and sphygmomanometers in health care, including in the context of the Minamata Convention on Mercury: key considerations and step-by-step guidance *(62)*

WHO 2011: A step-by-step guide for phasing our mercury thermometers and sphygmomanometers *(63).*

Selected tools
Highly hazardous pesticides HHPs WHO 2020: *The WHO recommended classification of pesticides by hazard and guidelines to classification (64)* FAO/WHO 2019: *Detoxifying agriculture and health from highly hazardous pesticides – a call for action (60)* FAO 2019: Pesticide Registration Toolkit *(45)* The FAO Pesticide Registration Toolkit provides technical advice on various processes and methods in pesticide registration, such as data requirements, assessment methods for parts of the registration dossier, decision-making steps, etc. **Other tools and resources can be found in the IOMC Toolbox.**
IOMC 2020: *IOMC Toolbox for decision making in chemicals management (19)* The IOMC Toolbox is a web-based platform that provides access to information and tools on the sound management of chemicals developed by participating organizations of the Inter-Agency Programme for the Sound Management Chemicals (IOMC), that is, FAO, the International Labour Organization (ILO), the United Nations Development Programme (UNDP), UNEP, the United Nations Industrial Development Organization (UNIDO), United Nations Institute for Training and Research (UNITAR), WHO, the World Bank and the Organisation for Economic Co-operation and Development (see also https://www.who.int/iomc/en/).
UNEP 2019: *Global chemicals outlook II – from legacies to innovative solutions (3)* SAICM 2021: *Chemicals Without Concern (66)*
UNEP 2019: Factsheet titled *Suggested steps for establishing a lead paint law (67)*
UNEP 2015: *UNEP guidance: on the development of legal and institutional infrastructures and measures for recovering costs of national administration for sound management of chemicals (68)*
UNEP 2019: *UNEP guidance – enforcement of chemicals control legislation (69)*
WHO 2020: *INCHEM database (13).* This database contains detailed information on the physico-chemical properties and toxicology of numerous chemicals.
WHO 2020: *Guidelines for establishing a poison centre (65)*
WHO 2015: *Health in All Policies: training manual (70)* This training resource guides the systematic consideration of health implications in all public policies to improve population health and health equity.
FAO 2000: *Assessing soil contamination. A reference manual (71)*
ILO/WHO 2020: *International Chemical Safety Cards (72)* ICSCs provide essential health and safety information on chemicals to promote their safe use. They are used at the “shop floor” level by workers, and by those responsible for health and safety in factories, agriculture, construction and other workplaces, and often form part of education and training activities. They can also be used by agencies responding to chemical incidents. ICSCs for more than 1700 chemicals are available in many languages.
ILO 2020: ILO instruments on the sound management of chemicals at work *(73)*
UNICEF/Pure Earth (2020): The toxic truth: children’s exposure to lead pollution is hindering a generation of potential *(74)*
UNICEF 2018: Understanding the impact of pesticides on children: a discussion paper *(75)*

5.3 Chemical incidents



This section focuses on incidents of chemical and radiological nature. The section *Selected tools* contains information relating to natural disasters and environmental health services during emergencies.

Overview

Chemical events arising from technological incidents, natural disasters, conflict and terrorism, polluted environments, and contaminated foods and products are common and occur worldwide. Between 2000 and 2020, there were over 1000 technological incidents involving chemicals worldwide, affecting over 1.85 million people *(76)*.

Some chemical incidents can have international consequences, for example when a chemically-contaminated product is distributed to multiple countries or when a chemical release contaminates an environmental medium such as air or water and subsequently traverses national borders *(77)*. It then falls under the IHR (2005) *(20)*. Under IHR (2005), Member States must have in place the necessary capacities to detect, evaluate and respond to public health events caused by any hazard, including chemicals. WHO, in turn, should provide assistance on request to Member States for investigating and controlling such events.

How do we assess a chemical (or radiological) incident?

Many chemical incidents are overt and are quickly recognized, such as a fire or large leak from a chemical plant. Some chemical releases may, however, only become apparent with the presentation or reporting of a number of cases with similar signs and symptoms, with common histories and linked in time and space. The timely identification of the cause of clusters or suspected outbreaks associated with exposure to chemicals may require a detailed investigation involving clinical, toxicological, epidemiological, environmental and laboratory analytical approaches.

Provided they are adequately resourced, poison centres can play a key role in identifying chemical incidents, and in supporting assessment and response. They are centres of expertise on clinical toxicology and have access to databases on products and substances. Most poison centres perform toxicovigilance, that is, they are engaged in the active process of identifying and assessing the toxic risks from exposure in a community or population to consumer products, pesticides, pharmaceuticals, environmental and industrial chemicals, controlled substances and natural toxins. Toxicovigilance involves the monitoring of data from poison centres to identify trends in poisoning exposures and the emergence of new risks associated with toxic substances *(65, 78)*.

What do we want to achieve?	
	Comprehensive management of chemical incidents requires prevention and preparedness, early detection and effective response and recovery. **Prevention** focuses on general measures that can be taken to diminish the likelihood of a chemical incident and to limit its severity. **Emergency planning and preparedness** details broad goals that can be accomplished to ensure adequate public health preparedness of all involved parties to respond to a chemical incident. **Detection and alert** describes various channels that can be used to detect a chemical incident and to alter the response of stakeholders involved in a chemical event emergency. **Response** deals with the public health tasks that should be carried out during an emergency. **Recovery** details the methods used to evaluate the causes and responses to chemical incidents and to follow up the victims in order to learn from the experience of incidents and near incidents, and to restore and remediate the affected environment *(79, 80)*. Acute chemical exposure guidelines are available from countries, including Acute Exposure Guideline Levels and the Immediately Dangerous To Life or Health values.

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
Policies and actions			
1. Implement international agreements into national laws. Selected international agreements include: • IHR (2005) – a legally binding agreement providing a framework to better prevent, prepare for and respond to public health events and emergencies of potential international concern, including chemical events *(20)*. • The ILO Prevention of Major Industrial Accidents Convention (C174) *(81)*.	Environment Health Other sectors	National	Regulation
2. Develop or update national policies and plans for prevention, preparedness, response, detection and recovery for chemical incidents, including for chemical incidents arising from natural hazard events (e.g. earthquakes, floods and cyclones) *(80)*.	Health Environment Other sectors	National	Regulation
Core capacities required under the International Health Regulations (2005) *(20, 77)*			
3. Establish designated focal points for IHR (2005) in all authorities that have an important role in the management of chemical events, for coordination and communication; establish a multisectoral national chemical emergency coordinating body; ensure adequate capacity for health-sector preparedness for prompt and adequate response to chemical events.	Health Environment	National	Governance

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
4. Implement a tested surveillance system for the detection, verification and risk assessment of chemical events of (potential) international health concern as part of a multi-hazard surveillance strategy and accompanied by a surveillance plan. Important sources of chemical incident notification and alert include: • poison centres; • hospital emergency departments; • primary health care facilities; • toxicology laboratories; • non-health sector sources such as agencies for consumer protection and food safety, environmental agencies, chemical plant operators, first responders and the public.	Environment Health	National	Assessment and surveillance
5. Implement tested emergency response plans taking into account possible event scenarios, addressing priority chemicals, hazardous sites and vulnerable populations. (Detailed information for the development of an emergency response plan is provided in the *Manual for the public health management of chemical incidents (80)*).	Environment Health	National	Other management and control
6. Ensure access to expertise, that is, maintaining an updated list and roster of experts and specialized centres, including poison centres, for: • risk assessment • exposure modelling • chemical fate and transport assessment • biological and environmental monitoring • (clinical) toxicology • diagnosis and treatment • health surveillance.	Environment Health	National	Information, education and communication
7. Ensure access to specialized drugs and equipment to be used by experts and/or specialized centres and to be placed strategically to ensure national coverage, including: • antidotes • PPE • decontamination equipment • equipment for biological and environmental monitoring.	Health Environment	National	Infrastructure, technology and built environment
8. Ensure access to toxicological and environmental laboratories, that is, laboratories are prepared to accept and analyse human and environmental samples at the time of a chemical emergency and arrangements are in place to ship the samples.	Health Environment	National	Information, education and communication; other management and control
9. Conduct chemical event scenario analysis including the modelling of adverse impacts for guiding the building of surveillance and response plans and to develop related capacities.	Environment Health	National	Other management and control

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
Additional recommendations for prevention, preparedness, detection, response and recovery of chemical incidents			
Prevention			
10. Avoid locating chemical facilities in hazard-prone or densely populated areas.	Land use planning	National	Regulation
11. Enforce a minimum set of safety standards and building regulations for all chemical facilities.	Environment	National	Regulation
12. Restrict and control chemical transportation and storage, including the requirement of licensing of hazardous sites and transport routes.	Environment	National	Regulation
13. Implement labour health and safety regulations including minimum levels of training, chemical protection and medical surveillance.	Labour	National	Regulation
14. Control waste disposal sites.	Environment	National	Regulation
15. Implement inspections of hazardous sites and transportation.	Environment	National	Regulation
16. Implement early-warning systems for weather-related natural events.	Health Environment	National	Regulation
17. Raise awareness about potential exposures, vulnerabilities to and health impacts from chemicals.	Health	National; community Universal health coverage	Information, education and communication
Preparedness			
18. Establish databases on hazardous sites, contents of transportation, chemical information, health care resources and emergency contact information.	Environment Health	National	Assessment and surveillance
19. Implement an incident management system, that is, a standardized approach to the command, control and coordination of emergency response.	Environment Health	National	Other management and control
Response			
20. Stop the release of the chemical, prevent the spread of contamination and limit exposure.	Industry Environment Health Other sectors	National; community Universal health coverage	Other management and control

Guidance	Sector principally involved in planning/ implementation	Level of implementation	Instruments
21. Provide an initial risk assessment and advise and alert health care services.	Health Environment	National; community Universal health coverage	Assessment and surveillance
22. Disseminate information and advice to responders, the public and the media.	Health Environment	National; community Universal health coverage	Information, education and communication
23. Register all individuals exposed during the incident. Collect appropriate human and environmental samples (this may include blood, urine, soil and water samples).	Health Environment	National; community Universal health coverage	Assessment and surveillance
24. Conduct investigations during the incident.	Health Environment	National; community Universal health coverage	Assessment and surveillance
Recovery			
25. Provide victim support such as medical care and a single point of contact for information and advice.	Health	Health care; national; community Universal health coverage	Information, education and communication; other management and control
26. Register exposed persons for follow-up and surveillance.	Health	Health care Universal health coverage	Assessment and surveillance
27. Conduct risk and health outcome assessments and environmental assessments.	Health Environment	Health care; national; community Universal health coverage	Assessment and surveillance
28. Implement rehabilitation actions including remediation and restoration of the environment, actions to prevent a further occurrence through causative factor analysis and emergency response evaluation, and to improve the affected community's health.	Health Environment	Community; national Universal health coverage	Other management and control
29. Contribute to the information of the international community.	Health Environment	National	Information, education and communication

Selected tools
Chemical incidents
WHO 2018: *Chemical releases caused by natural hazard events and disasters. Information for public health authorities (82)* contains a guide to important resources and tools for chemical incidents and emergencies in general (Annex D).
WHO 2015: *International Health Regulations (2005) and chemical events (77)*
Natural incidents
WHO 2018: *Chemical releases caused by natural hazard events and disasters. Information for public health authorities (82)*
WHO Regional Office for Europe 2017: *Flooding: managing health risks in the WHO European Region (83)*
General
WHO 2019: *Health Emergency and Disaster Risk Management Framework (79)*
WHO 2002: *Environmental health in emergencies and disasters: a practical guide (84)*

References

1. American Chemistry Society. CAS Registry. 2020 (https://www.cas.org/support/documentation/chemical-substances, accessed 23 December 2020).
2. International Council of Chemical Associations, UNEP. Debunking the Myths: Are There More than 100,000 Chemicals in Commerce? . International Council of Chemical Associations; 2020 (https://icca-chem.org/wp-content/uploads/2020/05/ICCA_DataAvailabilityStudy_Infographic.pdf, accessed 23 December 2020).
3. Global chemicals outlook II – from legacies to innovative solutions. Nairobi: United Nations Environment Programme; 2019 (https://www.unenvironment.org/resources/report/global-chemicals-outlook-ii-legacies-innovative-solutions, accessed 15 January 2021).
4. Chemicals and gender: United Nations Development Programme; 2011 (https://www.undp.org/content/undp/en/home/librarypage/environment-energy/chemicals_management/chemicals-and-gender.html, accessed 15 January 2021).
5. Chemicals road map and workbook. Geneva: World Health Organization; 2017 (https://www.who.int/publications/i/item/WHO-FWC-PHE-EPE-17.03, accessed 15 June 2021).
6. Globally harmonized system of classification and labelling of chemicals (GHS). New York and Geneva: UN; 2011 (https://www.unece.org/fileadmin/DAM/trans/danger/publi/ghs/ghs_rev04/English/ST-SG-AC10-30-Rev4e.pdf).
7. The Public Health Impacts of Chemicals: Knowns and Unknowns (incl. Data addendum for 2016) Geneva: World Health Organization; 2016 (http://www.who.int/ipcs/publications/chemicals-public-health-impact/en/, accessed 15 January 2021).
8. Joint FAO/WHO Expert Committee on Food Additives (JECFA). Geneva: World Health Organization; 2020 (http://www.fao.org/food-safety/scientific-advice/jecfa/en/#, accessed 15 June 2021).
9. Human biomonitoring: facts and figures. Copenhagen. Copenhagen: WHO Regional Office for Europe; 2015 (https://apps.who.int/iris/handle/10665/164588, accessed 18 May 2020).
10. Assessment of prenatal exposure to mercury: standard operating procedures Geneva: World Health Organization; 2018 (https://apps.who.int/iris/handle/10665/332161, accessed 28 July 2020).
11. Water safety planning: a roadmap to supporting resources. Geneva: World Health Organization; 2017 (https://www.who.int/water_sanitation_health/publications/wsp-roadmap.pdf?ua=1, accessed 27 March 2020).
12. Air quality guidelines – global update 2005. Particulate matter, ozone, nitrogen dioxide and sulfur dioxide. Copenhagen: WHO Regional Office for Europe; 2006 (https://apps.who.int/iris/handle/10665/107364, accessed 28 June 2021).
13. INCHEM. Geneva: World Health Organization; 2020 (http://www.inchem.org/pages/about.html, accessed 20 February 2020).
14. WHO, FAO. Joint FAO/WHO Meeting on Pesticide Residues (JMPR). Geneva: World Health Organization; 2020 (http://www.fao.org/agriculture/crops/thematic-sitemap/theme/pests/jmpr/en/, accessed 15 June 2021).
15. FAO, WHO. Codex Alimentarious. Rome: Food and Agriculture Organization; 2020 (http://www.fao.org/fao-who-codexalimentarius/home/en/, accessed 20 February 2020).
16. WHO guidelines for indoor air quality - selected pollutants. Copenhagen: WHO Regional Office for Europe; 2010 (https://apps.who.int/iris/handle/10665/260127, accessed 7 October 2019).
17. WHO, FAO. Food safety collaborative platform. Geneva: World Health Organization; 2020 (https://apps.who.int/foscollab/Dashboard/FoodConta, accessed

18. Food safety databases. Geneva: World Health Organization; 2020 (https://www.who.int/teams/nutrition-and-food-safety/databases, accessed 15 June 2021).
19. IOMC toolbox for decision-making in chemicals management. Geneva: Inter-Agency Programme for the Sound Management Chemicals; 2020 (https://www.IOMCToolbox.org, accessed 15 June 2021).
20. International Health Regulations (2005). 3 ed. Geneva: World Health Organization; 2016 (https://www.who.int/ihr/publications/9789241580496/en/, accessed 22 July 2020).
21. Minamata Convention on Mercury. UN Environment; 2013 (http://www.mercuryconvention.org/Convention/Text/tabid/3426/language/en-US/Default.aspx, accessed 30 September 2018).
22. Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal. UN Environment Programme; 1989 (http://www.basel.int/Home/tabid/2202/Default.aspx, accessed 21 February 2020).
23. Rotterdam Convention on the Prior Informed Consent Procedure for Certain Hazardous Chemicals and Pesticides in International Trade. UN Environment Programme; 2017 (http://www.pic.int/TheConvention/Overview/TextoftheConvention/tabid/1048/language/en-US/Default.aspx, accessed 21 February 2020).
24. Stockholm Convention on persistent organic pollutants (POPs) UN Environment Programme; 2017 (http://chm.pops.int/, accessed 21 February 2020).
25. Montreal Protocol on substances that deplete the ozone layer. 1998 assessment report of the technology and economic assessment panel. Nairobi: UN Environment Programme; 1998 (https://wedocs.unep.org/bitstream/handle/20.500.11822/8236/-Environmental%20Effects%20of%20Ozone%20Depletion_%201998%20Assessment-19981912.pdf?sequence=2&amp%3BisAllowed=, accessed 15 January 2021).
26. WHO, UNEP. Healthy environments for healthy children: key messages for action2010 (https://apps.who.int/iris/handle/10665/44381).
27. Inheriting a sustainable world? Atlas on children's health and the environment. Geneva: World Health Organization; 2017 (https://apps.who.int/iris/handle/10665/254677, accessed 15 June 2021).
28. Guidelines on the prevention of toxic exposures. Geneva: World Health Organization; 2004 (https://apps.who.int/iris/handle/10665/42714, accessed 15 January 2021).
29. International lead poisoning prevention week of action, 21–27 October 2018. Geneva: World Health Organization; 2018 (https://www.who.int/news-room/events/detail/2018/10/21/default-calendar/international-lead-poisoning-prevention-week-of-action, accessed 15 June 2021).
30. World directory of poison centres Geneva: World Health Organization; 2020 (https://apps.who.int/poisoncentres/, accessed 29 July 2020).
31. Exposure to arsenic: A major public health concern. Geneva: World Health Organization; 2019 (https://apps.who.int/iris/handle/10665/329482, accessed 13 May 2020).
32. Asbestos. Geneva: World Health Organization; 2020 (https://www.who.int/news-room/fact-sheets/detail/asbestos-elimination-of-asbestos-related-diseases, accessed 15 June 2021).
33. Exposure to lead: A major public health concern. Geneva: World Health Organization; 2019 (https://apps.who.int/iris/handle/10665/329953, accessed 13 May 2020).
34. Exposure to cadmium: A major public health concern. Geneva: World Health Organization; 2019 (https://apps.who.int/iris/handle/10665/329480, accessed 13 May 2020).
35. Exposure to dioxins and dioxin-like substances: A major public health concern. Geneva: World Health Organization; 2019 (https://apps.who.int/iris/handle/10665/329485, accessed 13 May 2020).
36. FAO, WHO. Code of practice for the prevention and reduction of dioxins, dioxin-like PCBs and non-dioxin-like PCBs in food and feed. Rome: Food and Agriculture Organization; 2018 (http://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXC%2B62-2006%252FCXC_062e.pdf, accessed 13 May 2020).
37. Inadequate or excess fluoride: a major public health concern. Geneva: World Health Organization; 2019 (https://apps.who.int/iris/handle/10665/329484, accessed 7 October 2020).
38. Exposure to highly hazardous pesticides: A major public health concern. Geneva: World Health Organization; 2019 (https://apps.who.int/iris/handle/10665/329501, accessed 13 May 2020).
39. Recycling used lead-acid batteries: health considerations. Geneva: World Health Organization; 2017 (https://apps.who.int/iris/bitstream/handle/10665/259447/9789241512855-eng.pdf, accessed 15 June 2021).
40. Children and digital dumpsites: E-waste exposures and health. Geneva: World Health Organization; 2021.
41. IOMC, WHO. Brief guide to analytical methods for measuring lead in blood. Geneva: World Health Organization; 2011 (https://apps.who.int/iris/handle/10665/77912, accessed 1 June 2021).
42. Public health impacts of exposure to mercury and mercury compounds: the role of WHO and ministries of public health in the implementation of the Minamata Convention. Geneva: World Health Organization; 2014 (https://apps.who.int/gb/ebwha/pdf_files/WHA67/A67_R11-en.pdf?ua=1, accessed 13 May 2020).
43. Minamata Convention on Mercury: annotated bibliography of WHO information. Geneva: World Health Organization; 2021 (https://www.who.int/publications/i/item/9789240022638, accessed 1 June 2021).
44. FAO, WHO. International Code of Conduct on Pesticide Management. Guidelines on Pesticide Legislation. Rome: Food and Agriculture Organization; 2015 (https://apps.who.int/iris/handle/10665/195648, accessed 13 May 2020).
45. FAO. Pesticide Registration Toolkit. Rome: Food and Agriculture Organization; 2019 (http://www.fao.org/pesticide-registration-toolkit/en/, accessed 13 May 2020).

46. OECD, FAO. OECD-FAO Guidance for Responsible Agricultural Supply Chains. Paris: OECD; 2016 (https://mneguidelines.oecd.org/OECD-FAO-Guidance.pdf, accessed 14 May 2020).
47. FAO, WHO. Managing pesticides in agriculture and public health - An overview of FAO and WHO guidelines and other resources. Rome: Food and Agriculture Organization; 2019 (http://www.fao.org/3/ca5201en/ca5201en.pdf, accessed 13 May 2020).
48. WHO, UNICEF. Arsenic primer: guidance on the investigation & mitigation of arsenic contamination. New York (NY): United Nations Children's Fund; 2018 (https://www.unicef.org/documents/arsenic-primer-guidance-investigation-mitigation-arsenic-contamination, accessed 4 June 2021).
49. Chrysotile asbestos. Geneva: World Health Organization; 2014 (http://apps.who.int/iris/handle/10665/143649/, accessed 16 October 2018).
50. UNEP, Stockholm Convention. Toolkit for identification and quantification of releases of dioxins, furans and other unintentional POPs under Article 5 of the Stockholm Convention. United Nations Environment Programme; 2013 (https://webdosya.csb.gov.tr/db/pops/editordosya/UNEP-POPS-GUID-2013-Toolkit%20PCDDF-En.pdf, accessed 1 October 2018).
51. Oral health surveys. Basic methods. 5th ed. Geneva: World Health Organization; 2013 (https://apps.who.int/iris/handle/10665/97035, accessed 8 October 2020).
52. Global elimination of lead paint: why and how countries should take action: policy brief. Geneva: World Health Organization; 2020 (https://apps.who.int/iris/handle/10665/333812, accessed 8 October 2020).
53. Global elimination of lead paint: why and how countries should take action: technical brief. Geneva: World Health Organization; 2020 (https://apps.who.int/iris/handle/10665/333840, accessed 8 October 2020).
54. Brief guide to analytical methods for measuring lead in paint. 2nd ed. Geneva: World Health Organization; 2020 (https://www.who.int/publications/i/item/9789240006058, accessed 15 January 2021).
55. Brief guide to analytical methods for measuring lead in blood. 2nd ed. Geneva: World Health Organization; 2020 (https://apps.who.int/iris/handle/10665/333914/, accessed 15 January 2021).
56. Guidance on organizing an advocacy or awareness-raising campaign on lead paint. Geneva: World Health Organization; 2020 (https://www.who.int/publications/i/item/9789240011496, accessed 15 June 2021).
57. Model Law and Guidance for Regulating Lead Paint. Nairobi: UN Environment Programme; 2018 (https://www.unenvironment.org/resources/publication/model-law-and-guidance-regulating-lead-paint, accessed 6 May 2020).
58. Exposure to mercury: a major public health concern, second edition. Geneva: World Health Organization; 2021 (https://www.who.int/publications/i/item/9789240023567, accessed 1 June 2021).
59. Addressing health when developing national action plans on artisanal and small-scale gold mining under the Minamata Convention on Mercury Geneva: World Health Organization; 2019 (https://www.who.int/publications/i/item/WHO-CED-PHE-EPE-19.9, accessed 15 June 2021).
60. FAO, WHO. Detoxifying agriculture and health from highly hazardous pesticides — a call for action. Rome: Food and Agriculture Organization; 2019 (http://www.fao.org/publications/card/en/c/CA6847EN/, accessed 15 June 2021).
61. Health sector involvement in the Minamata convention on mercury: outcomes of World Health Organization regional workshops for ministries of health. Geneva: World Health Organization; 2018 (https://apps.who.int/iris/handle/10665/275938, accessed 15 June 2021).
62. Developing national strategies for phasing out mercury-containing thermometers and sphygmomanometers in health care, including in the context of the Minamata Convention on Mercury: key considerations and step-by-step guidance. Geneva: World Health Organization; 2015 (https://apps.who.int/iris/handle/10665/259448, accessed 15 January 2021).
63. Replacement of mercury thermometers and sphygmomanometers in health care. Geneva: World Health Organization; 2011 (http://apps.who.int/iris/handle/10665/44592, accessed 15 June 2021).
64 WHO recommended classification of pesticides by hazard and guidelines to classification, 2019 edition. Geneva: World Health Organization; 2020 (https://apps.who.int/iris/handle/10665/332193, accessed 15 August 2021).
65. Guidelines for establishing a poison centre. Geneva: World Health Organization; 2020 (https://www.who.int/publications/i/item/9789240009523).
66. Chemicals Without Concern. In: Knowledge [website]. Strategic Approach to International Chemicals Management; 2021 (https://chemicalswithoutconcern.org/project/chemicals-without-concern, accessed 27 June 2021).
67. Suggested steps for establishing a lead paint law. Factsheet. . Nairobi: United Nations Environment Programme; 2019 (https://www.unenvironment.org/resources/factsheet/suggested-steps-establishing-lead-paint-law, accessed 15 January 2021).
68. UNEP Guidance: On the development of legal and institutional infrastructures and measures for recovering costs of national administration for sound management of chemicals Nairobi: UN Environment Programme; 2015 (https://wedocs.unep.org/handle/20.500.11822/12224, accessed 15 January 2021).
69. UNEP Guidance - Enforcement of Chemicals Control Legislation. Nairobi: UN Environment Programme; 2019 (https://wedocs.unep.org/handle/20.500.11822/28402, accessed 15 January 2021).
70. Health in All Policies: training manual. Geneva: World Health Organization; 2015 (https://apps.who.int/iris/handle/10665/151788, accessed 15 June 2021).

71. FAO. Assessing soil contamination. A reference manual. Rome: Food and Agriculture Organization; 2000 (http://www.fao.org/3/X2570E/X2570E00.htm, accessed 6 May 2020).
72. International chemical safety cards. Geneva: World Health Organization; 2020 (https://www.who.int/ipcs/publications/icsc/icsc_leaflet_en.pdf, accessed 15 June 2021).
73. The sound management of chemicals and waste in the world of work. Geneva: International Labour Organization; 2020 (https://www.ilo.org/wcmsp5/groups/public/---ed_protect/---protrav/---safework/documents/publication/wcms_731974.pdf, accessed 1 June 2021).
74. UNICEF, Pure Earth. The Toxic Truth: Children's exposure to lead pollution is hindering a generation of potential. New York: UNICEF; 2020 (https://www.unicef.org/media/73246/file/The-toxic-truth-children%E2%80%99s-exposure-to-lead-pollution-2020.pdf, accessed 15 January 2021).
75. Understanding the impact of pesticides on children: a discussion paper. New York: UNICEF; 2018 (https://www.unicef.org/csr/files/Understanding_the_impact_of_pesticides_on_children-_Jan_2018.pdf, accessed 15 January 2021).
76. Centre for Research on the Epidemiology of Disasters. Emergency Events Database (EM-DAT). 2018 (https://www.emdat.be/, accessed 15 June 2021).
77. International Health Regulations (2005) and chemical events. Geneva: World Health Organization; 2015 (https://ahpsr.who.int/publications/i/item/international-health-regulations-(2005)-and-chemical-events, accessed 8 June 2021).
78. WHO, IPCS. Guidelines for poison control. Geneva: World Health Organization; 1997 (https://apps.who.int/iris/handle/10665/41966, accessed 22 July 2020).
79. Health Emergency and Disaster Risk Management Framework. Geneva: World Health Organization; 2019 (https://apps.who.int/iris/handle/10665/326106, accessed 8 June 2021).
80. Manual for the public health management of chemical incidents. Geneva: World Health Organization; 2009 (https://apps.who.int/iris/handle/10665/44127, accessed 15 June 2021).
81. C174 - Prevention of Major Industrial Accidents Convention, 1993 (No. 174). Geneva: International Labour Organization; 1993 (http://www.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:55:0::NO::P55_TYPE,P55_LANG,P55_DOCUMENT,P55_NODE:CON,en,C174,/Document, accessed 22 July 2020).
82. Chemical releases caused by natural hazard events and disasters – information for public health authorities. Geneva: World Health Organization; 2018 (https://apps.who.int/iris/handle/10665/272390, accessed 23 July 2020).
83. Flooding: Managing health risks in the WHO European Region. Copenhagen: WHO Regional Office for Europe; 2017 (https://apps.who.int/iris/handle/10665/329518, accessed 15 January 2021).
84. Environmental health in emergencies and disasters: a practical guide. Geneva: World Health Organization; 2002 (https://apps.who.int/iris/handle/10665/42561, accessed 15 January 2021).

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